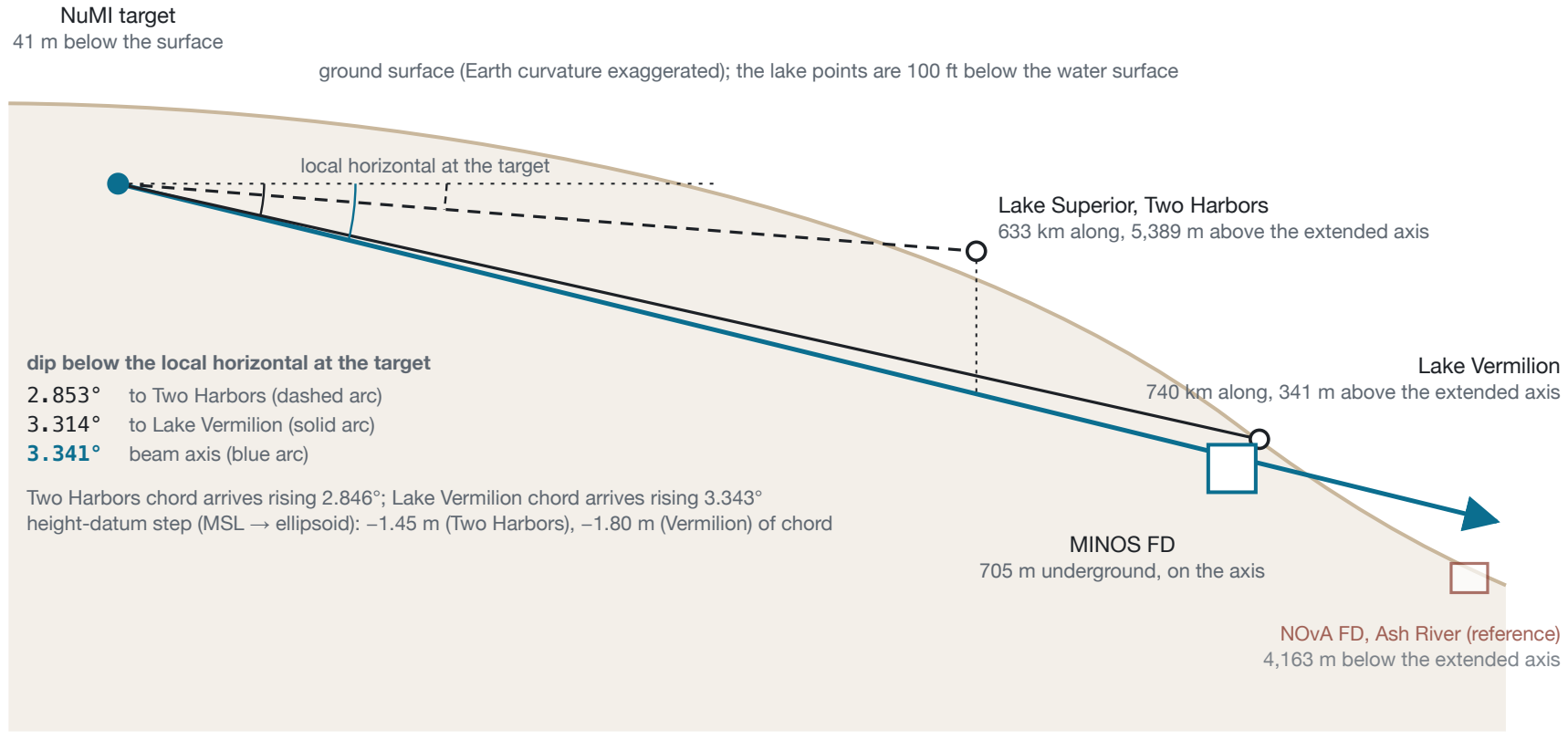
NuMI target
41 m below the surface
ground surface (Earth curvature exaggerated); the lake points are 100 ft below the water surface
local horizontal at the target
Lake Superior, Two Harbors
633 km along, 5,389 m above the extended axis
Lake Vermilion
740 km along, 341 m above the extended axis
dip below the local horizontal at the target
2.853° to Two Harbors (dashed arc)
3.314° to Lake Vermilion (solid arc)
3.341° beam axis (blue arc)
Two Harbors chord arrives rising 2.846°; Lake Vermilion chord arrives rising 3.343°
height-datum step (MSL → ellipsoid): −1.45 m (Two Harbors), −1.80 m (Vermilion) of chord
MINOS FD
705 m underground, on the axis
NOvA FD, Ash River (reference)
4,163 m below the extended axis